

A proposal to host the 2030 IEEE International Midwest Symposium on Circuits and Systems (MWSCAS) in Buffalo, NY, USA

August 4th – 7th, 2030

General Co-Chairs: Bibhu Datta Sahoo and Jeffery Walling

Symposium Theme: TBD

Introduction:

We are excited to propose hosting the 2030 IEEE International Midwest Symposium on Circuits and Systems (MWSCAS) in Buffalo, home to the University at Buffalo and the second largest city in New York. The selection of Buffalo Niagara as the host city for the IEEE Midwest Symposium on Circuits and Systems (MWSCAS) presents a compelling opportunity, especially given the conference has not been held in New York state since 1987.

Buffalo is a highly accessible city, ideal for hosting events like the IEEE MWSCAS. While not in the Midwest, its location provides convenient access to a variety of metropolitan areas and is well-connected by air.

The city is within a reasonable driving distance of several major U.S. and Canadian cities. To the east, attendees can easily drive from **Rochester** and **Syracuse**, while to the south, **Cleveland** and **Pittsburgh** are accessible by car. Most notably, Buffalo is just a short drive from **Toronto**, Canada's largest city, and the popular tourist destination of **Niagara Falls, Ontario**. This geographic positioning makes Buffalo a convenient hub for attendees traveling throughout the Northeast and Southern Ontario, offering easy road travel for a significant portion of the population.

For those traveling from further away, the **Buffalo Niagara International Airport (BUF)** provides excellent national and international connectivity. The airport offers direct flights to many major U.S. cities, including hubs like New York City, Chicago, and Atlanta. This makes it easy for attendees from across the country to reach the conference without a layover. The airport also offers some international routes and is a short drive from Toronto's Pearson International Airport (YYZ), which serves as a major international gateway with numerous flights from around the world. This combination of convenient regional driving access and direct air travel makes Buffalo a highly accessible and practical location for the IEEE MWSCAS, ensuring a wide range of attendees can easily participate.

Beyond its accessibility, the city provides a superb conference infrastructure to host a major conference like the IEEE MWSCAS, combining a large-scale convention center with a robust network of hotels and convenient transportation options. Buffalo-Niagara region also offers a wide range of unique and delicious food options which the attendees of IEEE MWSCAS will find plenty to savor beyond the conference halls. The iconic **Anchor Bar** is more than just a restaurant; it is a piece of culinary history. Famous as the birthplace of the **Buffalo Wings**, the bar holds a special place in the hearts of locals and food enthusiasts worldwide. Conference attendees can make a pilgrimage to the original location on Main Street, where the first wing was served back in 1964 and since then changed the bar menus across the globe. The atmosphere, steeped in nostalgia has license plates from all over the world covering the walls. While the menu offers a full selection of **Italian-American** dishes, the visit is incomplete without a basket of their signature deep-fried wings, served with celery and chunky blue cheese dressing—the authentic way.

Finally, the Buffalo Niagara region offers a unique "destination" appeal that will boost attendance. The world-famous Niagara Falls is just a 25-minute drive from downtown, providing a magnificent natural wonder for pre- or post-conference excursions.

In summary, Buffalo Niagara's blend of accessibility, affordability, and unique attractions makes it the perfect host to welcome IEEE MWSCAS back to New York state after almost 4-decades.

Personnel:

The organizing committee includes members from various U.S. and International Universities, National Laboratories, and major companies. The committee leaders have considerable experience in organizing IEEE conferences for the Circuits and Systems Society. The committee will also feature several people with limited prior experience in conference organization to ensure that there is a diverse pool of experienced individuals for organizing future conferences.

General Co-Chairs (MWSCAS):

Dr. Bibhu Datta Sahoo (University at Buffalo)
Dr. Jeffery Walling (Virginia Tech)

Technical Program Chairs:

Dr. Aritra Banerjee (University at Illinois Chicago)
Dr. Subhanshu Gupta (Ohio State University, Columbus)
Dr. James Stein (University of Nebraska, Lincoln)

Plenary Session Chairs:

Dr. Marilyn Wolf (University of Nebraska, Lincoln)
Dr. Jinjun Xiong (University of Texas, San Antonio)

Tutorial Chair:

Dr. Neeraj Magotra, (Western New England University)
Dr. Rajiv Joshi (IBM, Yorktown Heights)

Special Sessions Chair:

Dr. Wood Chiang (Brigham Young University, Provo)

Student Paper Contest Chair:

Dr. Chistian Chamon Garcia (Virginia Tech.)
Dr. Teju Das, (Rochester Institute of Technology)

Publications Chair: TBD

Local Arrangements Chair: TBD (From Buffalo)

Registration Chair: TBD

Web Page Coordinator: TBD

Wi-CAS Chair: TBD

Advisory Committee:

Dr. Randall L. Geiger (Iowa State University), Previous General Chairs

Dr. Bibhu Datta Sahoo (General Co-Chair, MWSCAS) joined the faculty of the University at Buffalo, Department of Electrical Engineering as Professor in Fall 2023. He received the B.Tech. degree in Electrical Engineering from the Indian Institute of Technology (IIT), Kharagpur, India, in 1998, the M.S.E.E. degree from the University of Minnesota, Minneapolis, MN, USA, in 2000, and the Ph.D. E.E. Degree from the University of California, Los Angeles in 2009. From 2000 to 2006, he was with DSP Microelectronics Group, **Broadcom Corporation, Irvine, CA**, where he designed analog and digital integrated circuits for signal-processing applications. From December 2008 to February 2010, he was with **Maxlinear Inc., Carlsbad, CA**, where he was involved in designing RF Integrated Circuits for CMOS TV tuners. He has been a faculty at IIT Kharagpur, India for 6 years and Amrita University, India for 5 years. His research interests include mixed signal circuit design, analog computing, and machine learning hardware. He received the 2008 Analog Devices Outstanding Student Designer Award and was the co-recipient of the 2013 CICC Best Paper Award. He was the Associate Editor of **IEEE Transactions on Circuits and Systems-II** from Aug. 2014 to Dec. 2015. Since Aug. 2019 he has been the Associate Editor of **IEEE Open Journal of Circuits and Systems**. He is one of the **IEEE Circuits and Systems Society (CASS) Distinguished Lecturer** for 2025-2026.

Dr. Jeffery Sean Walling (General Co-Chair, MWSCAS) received his B.S. degree from the University of South Florida (2000) and his M.S. and Ph.D. degrees from the **University of Washington** (2005, 2008). His career includes positions at **Motorola Solutions, Intel, UW, Rutgers University, the University of Utah, and Tyndall National Institute**. He is currently an Associate Professor at **Virginia Tech**.

He has authored more than 100 peer-reviewed articles. His research interests focus on mixed-signal RF, RF power amplifier design, and inverse design in electromagnetics.

Dr. Walling serves as an **Associate Editor** for **IEEE Transactions on Circuits and Systems—I & II** and has served on the Technical Program Committees for ISSCC and RFIC. In 2025, he was the North American Regional Chair of ISSCC, and he served as an IEEE Solid-State Circuits Society Distinguished Lecturer from 2023 to 2024.

His honors include the Analog Devices Outstanding Student Designer Award (2006), Intel Predoctoral Fellowship (2007–2008), UW Yang Award (2008), MobiCom Best Paper Award (2012), and teaching awards from Rutgers University (2012) and the University of Utah (2015).

Meeting Facilities:

There are many excellent hotels (3-star and above) and restaurants at affordable prices in the Buffalo-Niagara area. The city itself has all the amenities needed to host midsize to large conferences. We are considering three leading venues (described below) with on-site meeting and dining facilities and a long history of supporting successful conferences with 300-400 people (which is our expected attendance level for MWSCAS 2030). We plan to use IEEE Meetings, Conferences, and Events service to help negotiate the contract with the chosen venue.

Option 1 (Preferred): Buffalo Marriott Niagara:

The Grand Ballroom at this hotel can host up to 500 people in a classroom-style setup or 800 in a theater-style setup.

Option-2: Sheraton Niagara Falls:

The largest meeting space at this hotel has a capacity for up to 500 people.

Option-3: DoubleTree by Hilton Hotel Niagara Falls New York:

The River's Edge Ballroom can accommodate up to 500 people in a school room setup, 600 in a theater-style setup, or 1000 for a reception.

Option-4: Niagara Falls Marriott on the Falls:

The Oakes Ballroom can hold up to 1000 people in a theater-style setup, 480 in a schoolroom setup, and 648 for a banquet.

Technical Program:

A large space (hotel ballroom) capable of hosting more than 400 people will be available for the plenary and keynote presentations as well as the general sessions. A total of 8 breakout rooms will be used to support 6 parallel technical sessions. Each breakout room will accommodate between 30 and 40 participants. The venue will also include a suitably large room for the poster sessions and coffee breaks.

The first day of the conference (August 4th) will be devoted to Tutorials and Short Courses, with all other sessions held on the three remaining days (August 5th-7th). We plan to have about 72 oral presentations each day (= 4 papers/session x 6 parallel sessions/slot x 3 time slots), resulting in a total of 216 oral presentations over three days. We also anticipate having 2 poster sessions each day of the conference with 20 poster presentations per session, resulting in a total of 120 poster slots. Accordingly, we plan to accommodate a total of 336 papers for both oral and poster presentations. There will be a keynote address each day. Student paper contests for MWSCAS will be scheduled for the first day of the conference in parallel with the oral sessions. The technical paper review process will be handled using the ePapers system.

Special Sessions:

- TBD

Conference Schedule:

The conference will be held from August 4th – 7th, 2030. An outline of the proposed schedule is summarized below:

- Sunday August 4th: Tutorials and Short Courses, Welcome Reception
- Monday August 5th: Keynote 1, Oral and Poster Sessions
- Tuesday August 6th: Keynote 2, Oral and Poster Sessions, Conference Banquet
- Wednesday August 7th: Keynote 3, Oral and Poster Sessions, Award Ceremony

Transportation:

All four conference hotels being considered are in the Buffalo-Niagara area with major airports within reasonable driving distance:

- Buffalo Intl. Airport within 20 mins of driving distance
- Rochester Airport within 1 hr of driving distance

- Toronto International Airport within 2 hrs of driving distance. Various Airport shuttles from Toronto Airport to Buffalo-Niagara area are available. Rental car is another option. However, rental cars should have the permit to cross the Canada-US border.

Accommodations:

Buffalo is a major metropolitan area and has many excellent hotels (3-star and above) and restaurants at affordable prices. Thus, conference attendees will have no difficulty in finding accommodation to suit their tastes and budget. The following are a few hotels that can be considered:

Hotels in Buffalo, NY

- **Buffalo Marriott Niagara (Preferred):** The Grand Ballroom at this hotel can host up to 500 people in a classroom-style setup or 800 in a theater-style setup.
- **M Hotel Buffalo:** This hotel offers seven different event spaces, with the largest accommodating up to 500 guests.
- **500 Pearl / Aloft Buffalo Downtown:** This complex has multiple venues, including the Forbes Theater, which has a capacity of 400 for cocktails and 288 for banquets.

Hotels in Niagara Falls, NY

- **Sheraton Niagara Falls:** The largest meeting space at this hotel has a capacity for up to 500 people.
- **DoubleTree by Hilton Hotel Niagara Falls New York:** The River's Edge Ballroom can accommodate up to 500 people in a school room setup, 600 in a theater-style setup, or 1000 for a reception.
- **Niagara Falls Marriott on the Falls:** The Oakes Ballroom can hold up to 1000 people in a theater-style setup, 480 in a schoolroom setup, and 648 for a banquet.
- **Seneca Niagara Resort & Casino:** This resort has a total event space of 29,496 sq. ft., with its largest conference room at 24,443 sq. ft., which can accommodate a large number of guests.

Local Attractions:

The Buffalo and Niagara Falls region of New York offers a rich combination of natural wonders, historical sites, and cultural attractions. From the powerful cascades of Niagara Falls to Buffalo's revitalized waterfront and impressive architecture, there's a wide variety of activities for every type of traveler.

Niagara Falls Attractions

The most famous attraction in the area is, of course, Niagara Falls itself. The American side offers numerous ways to experience the falls up close.

- **Niagara Falls State Park:** America's oldest state park, it encompasses the American, Bridal Veil, and Horseshoe Falls. It's free to enter and offers stunning views from various vantage points.
- **Maid of the Mist:** This iconic boat tour takes you right into the basin of Horseshoe Falls, providing a truly immersive and wet experience of the falls' immense power.
- **Cave of the Winds:** Descend 175 feet into the Niagara Gorge and walk on a series of wooden decks at the base of the Bridal Veil Falls, getting incredibly close to the rushing water on the "Hurricane Deck."

- **Niagara Falls Observation Tower:** This tower offers the only panoramic view of all three waterfalls in the park, including a view of the Canadian side.
- **Aquarium of Niagara:** Just a few blocks from the state park, this aquarium is a great family-friendly option with a focus on aquatic life.
- **Old Fort Niagara:** Located a short drive north of the falls in Youngstown, this historic fort at the mouth of the Niagara River on Lake Ontario was a key military post in the 17th and 18th centuries and offers living history demonstrations.

Buffalo Attractions

A short drive from Niagara Falls, Buffalo has its own unique character and a host of attractions.

- **Canalside and the Waterfront:** Once the western terminus of the Erie Canal, this area has been revitalized into a bustling hub for recreation, dining, and events. In the summer, you can enjoy kayaking and concerts, while winter brings a massive ice-skating rink.
- **Architecture:** Buffalo is an architectural gem. Highlights include Frank Lloyd Wright's **Darwin D. Martin House**, considered one of his finest works, and the magnificent Art Deco **Buffalo City Hall**, which offers a free observation deck with stunning views.
- **Museums and Culture:**
 - **Buffalo AKG Art Museum:** An impressive collection of modern and contemporary art.
 - **Theodore Roosevelt Inaugural National Historic Site:** The preserved site where Theodore Roosevelt was sworn into office.
 - **Buffalo and Erie County Naval and Military Park:** The largest inland naval park in the US, where you can tour decommissioned ships.
 - **Buffalo RiverWorks:** A unique entertainment complex built into old grain silos, offering adventure activities like rock climbing and a ropes course.
- **Parks and Outdoors:** **Delaware Park** is a Frederick Law Olmsted-designed park that is home to the **Buffalo Zoo**, one of the oldest in the country. The **Buffalo and Erie County Botanical Gardens** is another beautiful spot, housed in a Victorian-style glass conservatory.

Conference Support and Preliminary Budget:

We expect the CAS Society and the MWSCAS Steering Committee to both provide seed money to host the MWSCAS portion of the conference, as in previous years. We will seek additional in-kind support from the University at Buffalo (one of the largest public research universities in NY and local companies. We will also seek in-kind support (in the form of publicity and volunteers) from the local IEEE chapter.

Commitments and Endorsements:

The organizing committee has had initial discussions with the Chair of the EE Departments at the University at Buffalo regarding this proposal. The Chair has expressed his full support for the proposed hosting of MWSCAS at Buffalo.

Preliminary Budget:

Our revenue model assumes **350** registrations covering various categories as listed below:

Expected revenue from “MWSCAS” attendees

Type of registration	Expected number	Expected revenue
Advance Registration for IEEE CAS Members (\$850)	70	\$59,500
Advance Registration for IEEE Members (\$950)	110	\$104,500
Advance Registration for IEEE Non-Members (\$995)	30	\$29,850
Advance Registration for IEEE Student Members (\$450)	55	\$24,750
Advanced Registration for IEEE Life Members (\$450)	15	\$6,750
Standard Registration for IEEE CAS Members (\$900)	15	\$13,500
Standard Registration for IEEE Members (\$1050)	15	\$15,750
Standard Registration for IEEE Non-Members (\$1100)	10	\$11,000
Standard Registration for IEEE Student Members (\$550)	10	\$5,500
Standard Registration for Student Non-Members (\$600)	20	\$12,000
Extra Paper Charges (\$200 per paper)	35	\$7,000
Extra Banquet Tickets (\$100 each)	10	\$1,000
Total		\$291,100

Estimated expenditure

Welcome reception (200 x \$70)	\$14,000.00
8 Breakouts (220 x 6 x \$20+120 x 2 x \$20)	\$31,200.00
Lunch (4 days) (900 x \$60)	\$54,000.00
Banquet (300 x \$80)	\$24,000.00
Steering Committee Dinner (40 x \$70)	\$2,800.00
Management Services Expenses	\$0.00
Conference Publications	\$8,000.00
AV Rental	\$40,000.00
Space Rental	\$0.00
Promotion	\$2,500.00
New Initiatives	\$2,500.00
Administration	\$14,000.00
Keynote Speakers	\$10,000.00
Gifts for Participants	\$12,000.00
ePapers	\$10,000.00
Tax and Gratuity	\$8,000.00
Total	\$233,000

Estimated surplus: \$58,100 (20%)